MobileNet-based Prediction of Preterm Births

Tooba Noor-UL-Ain
Department of Electronics Engineering,
University of Engineering and Technology, Taxila
Taxila, Pakistan
toobanoor7@gmail.com

Ammara Abbas
Department of Electronics Engineering,
University of Engineering and Technology, Taxila
Taxila, Pakistan
ammara.abbas68@gmail.com

Muhammad Umar Khan
Department of Electronics Engineering,
University of Engineering and Technology, Taxila
Taxila, Pakistan
sa.umarkhan@gmail.com

Muhammad Faraz
Department of Electronics Engineering,
University of Engineering and Technology, Taxila
Taxila, Pakistan
muhammad.faraz@uettaxila.edu.pk

Syed Zohaib Hassan Naqvi
Department of Electronics Engineering,
University of Engineering and Technology, Taxila
Taxila, Pakistan
zohaib.naqvi@uettaxila.edu.pk

Sumair Aziz
Department of Electronics Engineering,
University of Engineering and Technology, Taxila
Taxila, Pakistan
sumair.aziz@uettaxila.edu.pk

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***Abstract*—Nowadays preterm birth is a serious medical problem since it causes most perinatal deaths and puts people at risk for additional illnesses that could negatively impact their health. The early detection of such cases may prove to be very beneficial in raising society's overall health standards. In this work, uterine electromyography, also known as Electro Hysterography (EHG), is used to propose such an early diagnosis methodology. A 4th-order band-pass Butterworth filter with a frequency of 0.8 to 5 Hz for each channel is used to filter the raw EHG signals from the EHG channels. Deep Learning (DL) algorithm is used for this purpose. Then, we extract features using DL, and the Quadratic Discriminant validation classifies EHG signals quite reliably. With 100% sensitivity and 99.5% specificity, the suggested framework has a 99.8% accuracy rate for classifying term and preterm birth. So this method provides an early pregnancy observation, preserving the lives of infants, and enhancing mother and baby health. This technology will assist in day-to-day clinical work.**

Keywords---- Deep Learning, non-invasive, Electro hysterography (EHG), Butterworth filter, Premature birth, Deep Learning (DL).

I. INTRODUCTION

Premature births occur when a baby is born before 37 weeks of pregnancy, whereas normal births occur after 37 weeks [1]. In 2009, around 7.3%, 7.1%, and 7.2% of babies born in England and Wales arrived preterm [2]. Preventing preterm labor and delivery is a major reason pregnant women are admitted to the hospital in the United States. This is important since over 10% of the four million babies born each year are premature. Taking care of these premature babies in specialist hospital units (NICU) costs roughly $1500 each day, totalling more than $25.6 billion in healthcare costs for parents who give birth in the hospital [3].

According to [4] there are a few major causes of premature births (PTB):

1. The membrane that covers the fetus can rupture before delivery (PPROM).
2. Doctors may speed up the delivery if it is medically required for the mother's or the baby's health.
3. Unwanted contractions, often known as preterm labor (PTL), might result in an early birth.
4. Mom's health concerns, including infections, continuing diseases, reproductive system disorders, or previous cervix surgery.
5. Things in the mother's environment, such as drugs, smoking, or stress, can also have a role.
6. Pregnancy-related hazards, such as vaginal bleeding, urethral irritation, or a shorter cervix.

Premature births (PTB) are significant because they can reduce the quality of life and health for both the baby and the mother [5]. Babies delivered too soon may have respiratory difficulties [6,] be very small, and develop major health problems. This can potentially impact the growth of their oral tissues [7]. According to research, premature babies may require extra assistance in school, particularly throughout the first twelve years of their lives.

To anticipate preterm deliveries, researchers have been investigating the electrical activity of the muscles in the uterus, known as EHG. They employ machine learning (ML) and deep learning (DL) approaches, such as convolutional neural networks (CNNs), to predict whether a baby will be born on time or early. DL techniques, such as CNNs, have proven to be superior because they can learn key patterns from input data and solve problems that traditional ML methods cannot [9]. These techniques have also been applied to other types of medical pictures, such as X-rays and MRI scans [10].

EHG is an intriguing area because it entails placing sensors on a pregnant woman's belly to capture signals from her uterus muscles. This is used to assess electrical activity in the uterus, similar to a "heartbeat" [11]. It is critical to monitor these signals during pregnancy to determine whether contractions are normal or if there is a danger of preterm birth [12]. Scientists have been hard at work figuring out how to utilize EHG to forecast whether a baby will be born early or late, and they've made some headway.

II. LITERATURE REVIEW

Due to its possible negative consequences on infant mortality and long-term health outcomes, preterm delivery, which occurs before 37 weeks of gestation, is a serious concern for maternal and child health. Early detection of premature births can greatly enhance clinical care and patient outcomes. Researchers have studied many strategies throughout the

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years, and most recently, deep learning techniques have demonstrated amazing promise for improving prediction accuracy. This literature review seeks to present an overview of the work carried out by previous researchers in forecasting preterm deliveries using deep learning approaches and the developments made in this field.

To predict preterm deliveries, machine learning (ML) techniques have been frequently used. Traditional ML methods like logistic regression and decision trees were used by researchers like Alfirevic et al. [13] and Ibitoye et al. [14] to predict preterm births based on clinical risk factors. These methods, however, frequently fail to recognize intricate patterns in the data, which reduces the precision of their predictions.

The discipline of predictive analytics has undergone a revolution in recent years with the introduction of deep learning (DL). Deep neural networks (DNNs) give predictive models the ability to automatically learn complex properties from unstructured input. This opportunity to more accurately predict preterm births has been welcomed by researchers.

To predict preterm deliveries, researchers have used a variety of deep learning models. Concept-based Recurrent Neural Network (RNN) and A-RNN models were introduced by Rawan Alsaad et al. [15] that combined temporal data and code-level idea models. Their method revealed enhanced attention performance, demonstrating the potency of fusing various neural network topologies.

Convolutional neural networks (CNNs) were used in a noteworthy study by Shen-Guan Wu et al. [16] to analyze Electrohysterogram (EHG) signals and predict premature birth. This work demonstrated the ability of CNNs to analyze EHG signals and emphasize important aspects, despite some accuracy issues.

Additionally, Lili Chen and Huoyao Xu [17] created a Deep Neural Network (DNN) with stacked shallow autoencoders (AEs) for predicting preterm and term births from uterine records. By using wavelet entropy and sample entropy for feature reduction, their model's classification performance was improved.

Recently, advanced DL architectures like Deep Belief Networks (DBNs), Extreme Learning Machines (ELMs), and 1D CNNs have been used. With the help of these models, more precise predictions can be made because they show the capacity to uncover significant patterns from various data sources.

Preterm deliveries can be predicted with remarkable success using deep learning techniques. These models differ from conventional ML algorithms in that they may automatically identify pertinent features from complex data, as opposed to the latter which frequently call for explicit feature engineering. To increase prediction accuracy, researchers have effectively used a variety of DL techniques, including RNNs, CNNs, DBNs, and ELMs.

Deep learning problems still exist, including the need for significant volumes of data, model interpretability, and computing needs. To improve the accuracy and practicality of deep learning models for forecasting preterm deliveries, several issues must be resolved.

An important step forward in the study of mother and child health has been made with the use of deep learning algorithms to predict preterm births. Researchers have looked into several DL structures and methodologies, demonstrating their capacity to correctly predict preterm deliveries. Deep learning models will probably become more important as technology and methodology advance in the early detection and management of premature deliveries.

The procedure of using the generated model to forecast term and preterm cases is shown graphically in Fig. 1.

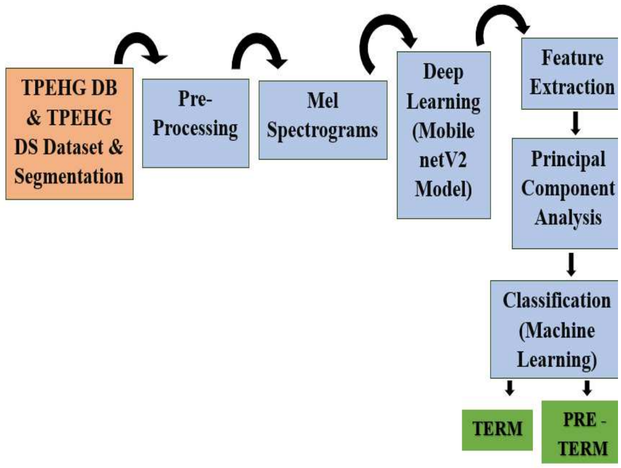

Fig. 1. System Block Diagram for prediction of term and preterm cases.

III. Materials and Methods

A. Data Acquisition and Segmentation

To extract unprocessed uterine Electrohysterogram (EHG) data, we used the PhysioNet platform and the WFDB Toolbox. There are 26 signals in the dataset, Term-Preterm Electro Hysterogram Dataset with Tocogram (TPEHGT DS), 13 from term pregnancies and 13 from preterm pregnancies. Additionally, we used the Term-Preterm EHG Database (TPEHG DB), which has 300 EHG records from 38 preterm participants and 262 term subjects. Both datasets have a signal length of 30 minutes and a sampling frequency of 20 Hz. Four bipolar electrodes, denoted E1, E2, E3, and E4, were placed strategically on the patients' bodies to monitor uterine activity and collect the EHG signals. Fig. 2 illustrates this arrangement.

These channels are determined as

1. Channel 1 = E2-E1
2. Channel 2 = E2- E3
3. Channel 3 = E4-E3

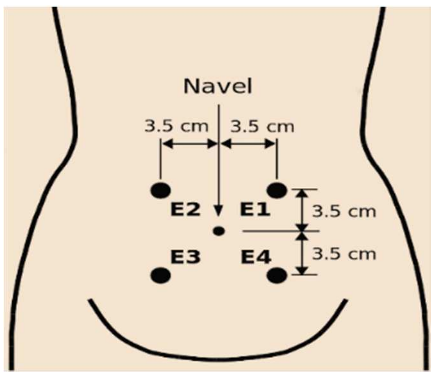

Fig. 2. Electrodes Position on Surface

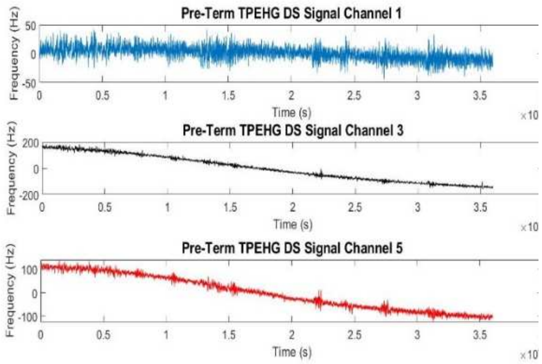

Fig. 3. Raw Preterm TPEHG DS Channel [1 3 5]

EHG signals were obtained from channels [1 3 5] of the TPEHG DS dataset and [1 5 9] channels of the TPEHG DB dataset. Then, we performed segmentation of the acquired EHG signal into a smaller signal of 4 secs duration. The frequency domain plot for TPEHG DS is shown in Fig. 3.

B. Pre Processing

The initial EHG signals are segmented into 4-minute segments during the preprocessing stage and have a total duration of 30 minutes that is further split into smaller partitions. Disruptive components like noise and motion artifacts are present in these signals by nature. A 4th order bandpass filter is used to narrow in on the specific focus areas. From 0.08 Hz to 5 Hz, this filter's frequency range effectively filters out unwanted noise and motion artifacts, particularly those caused by power lines. Several factors, such as motion artifacts and interference from the power lines, contribute to the noise in the EHG signals. Fig. 4 and 5, respectively, for each of the three channels after the filtering procedure was applied in TPEHG DB and TPEHG DS, show the frequency domain plots illustrative of the preterm EHG signals.

C. Mel spectrogram

The frequency content of Electrohysterogram (EHG) signals is represented by the Mel spectrograms, which are produced [20]. High-dimensional and redundant information are common characteristics of EHG signals. Mel spectrograms are used to provide a clear representation by lowering the signal's dimensionality while preserving its essential

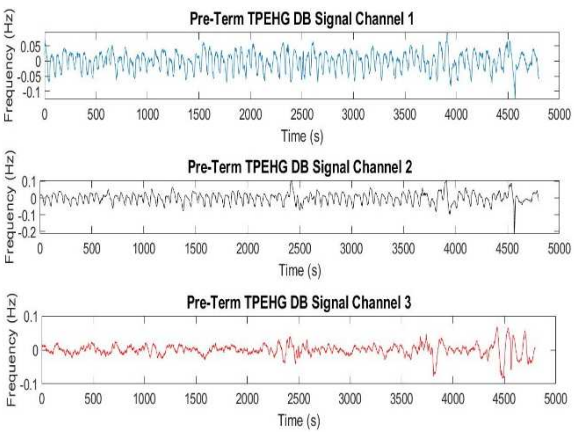

Fig. 4. Frequency Domain Plot for Pre-Term TPEHG DB Dataset after filtration

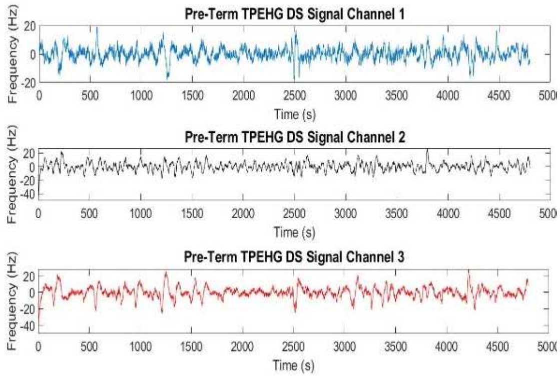

Fig. 5. Frequency Domain Plot for Pre-Term TPEHG DS Dataset after filtration

properties to address this. The EHG signals are processed by segmenting them into overlapping frames with a defined duration, frequently by using Hann windowing methods. The signal is transformed from the time domain to the frequency domain for each frame using the Fourier transform (FFT). On the magnitude spectrum that the Fourier transform yielded, a Mel filter bank is then applied. Logarithmic compression is applied to the magnitude values that emerge from the Mel filter bank. Fig. 6 shows a preterm signal's Mel spectrogram.

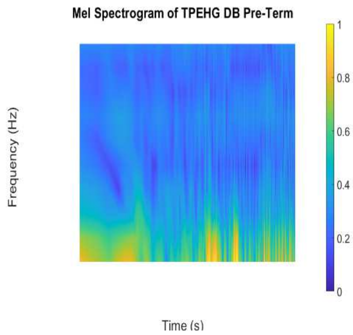

Fig. 6. Mel Spectrogram of Pre-Term signal

D. Features Extraction

Feature extraction is used to find out more precise results. In this part of the research we extract features of the given classes that have the specific which leads us to the more accurate prediction of true class On this basis, we predict whether the signal is of term and preterm class. For this purpose, we utilized mobilenetv2 architecture. We selected the CNN-based MobileNet-V2 deep learning model. The MobileNetV2 model architecture is a convolutional neural network (CNN) designed to be efficient and lightweight while maintaining superior performance on image classification tasks.

Architecture: MobileNetV2 shown in Fig 7 uses fully separable convolution as good components. It has two architectural elements,

- linear bottlenecks across the layers.
- shortcut connections between the bottlenecks.

MobileNetV2 has 53 convolution layers and 1 average pool. It is made up of two major parts:

- Residual block inverted.

- Residual bottleneck block

and uses two types of convolution layers:

- Convolution 1x1
- Depth-wise convolution 3x3

Out the total dataset, 70% is used for training, and 30% for testing. The architecture of MobileNetv2 is shown in Fig. 7.

We extract 1000 features from the Logit layer of mobilenetv2 which is the Fully connected layer of this model.

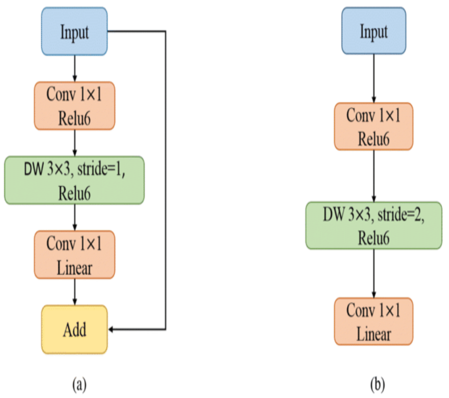

Fig. 7. a) 1st block of stride; b) 2nd block of stride

E. Data Balancing

Between the number of Term instances (262) and Preterm cases (38) in our sample, there is a substantial disparity. Synthetic Minority Oversampling Technique (SMOTE), which was used to correct this imbalance, was used. The non-majority class's number is boosted using this procedure, which entails the creation of false samples. In essence, SMOTE is used to generate additional instances of the underrepresented class, hence lowering the data discrepancy and assuring a more balanced representation of the classes [21]. When machine learning models are presented with unbalanced data, this can be especially useful in enhancing their performance.

F. Classification

We used a quadratic discriminant classifier to separate patients into term and preterm categories. Principle Component Analysis (PCA) was used as a step before classification once the features were extracted. This is a typical strategy where we use quadratic discriminant analysis to identify or separate the components inside the data. We produced accurate results in terms of classification accuracy by combining several kernels [22]. We can effectively distinguish between the two classes using this technique, and we can generate reliable predictions based on the properties of the data.

IV. Results and Discussion

Electrohysterogram (EHG) data have been used to build a signal-processing technique that makes it easier to distinguish between term and preterm babies. To identify the features that contain unique information, this method needs a thorough review of all available data. After the raw EHG signal has been preprocessed and segmented using a Deep Learning (DL) method, this analysis is carried out.

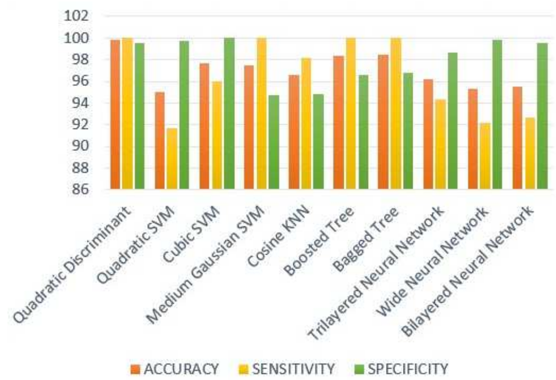

Fig. 8. Classifier Comparison

The metrics used to measure the performance of classifiers are accuracy, specificity, and sensitivity. The performance ratings of 10 are shown in Fig. 8 and Table 1.

TABLE 1. PERFORMANCE PARAMETERS OF EACH CLASSIFIER

MODEL	ACC	SEN	SPEC
Quadratic Discriminant	99.8	100	99.5
Quadratic SVM	95	91.7	99.7
Cubic SM	97.7	96	100
Medium Gaussian SVM	97.5	100	94.7
Cosine KNN	96.6	98.2	94.8
Boosted Tree	98.4	100	96.6
Bagged Tree	98.5	100	96.8
Trilayered Neural Network	96.2	94.3	98.7
Wide Neural Network	95.3	92.2	99.8
Bilayered Neural Network	95.5	92.7	99.5

In this study, a variety of classifiers were used, including the Quadratic Discriminant, Quadratic Support Vector Machine (SVM), Cubic SVM, Medium Gaussian SVM, Cosine k-nearest Neighbours (KNN), Boosted Tree, Bagged Tree, Trilayered Neural Network, Bilayered Neural Network, and Wide Neural Network. The quadratic discriminant was used to reach the maximum accuracy, which was a remarkable 99.8%. Additionally, the Cosine KNN demonstrated an accuracy of 96.6%, the Boosted Tree showed good results at roughly 98.4%, the Medium Gaussian SVM provided a performance of 97.5%, and the Cubic SVM produced an accuracy of 97.7%.

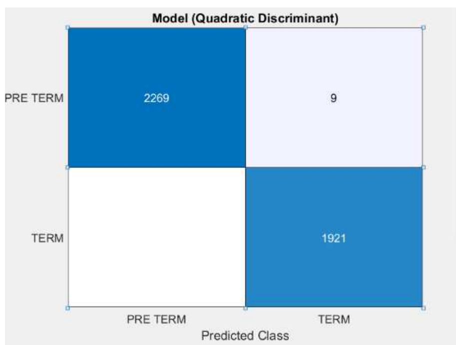

Fig. 9. Confusion Matrix Accuracy

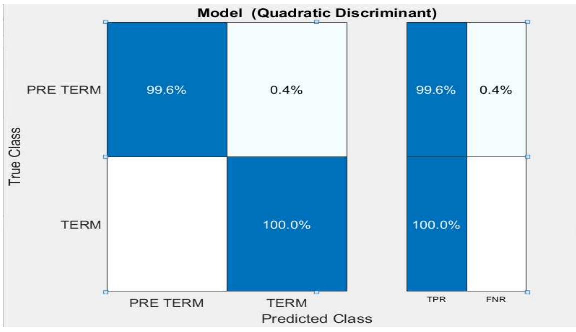

Fig. 10. Positive Rate Confusion Matrix

Class-wise performance parameters are shown in Fig. 9 and Fig. 10, which clearly show the Sensitivity is 100%, Specificity is 99.5% and Accuracy is overall 99.8% on Quadratic Discriminant.

V. CONCLUSION

In conclusion, our study offers a novel approach that, when compared to earlier studies, greatly improves the classification accuracy of term and preterm cases. We present a more straightforward technique that can be quickly adopted in hospitals and maternity institutions, and we introduce a different method for classifying EHG records of term and preterm pregnancies. The effectiveness of the methodology is unaffected by the reduced complexity. Another significant development is that it may help to enhance mother and newborn health outcomes. It has the potential to be extremely important in saving the lives of both new moms and their babies.

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